

Program : Diploma in Printing Technology	
Course Code : 4101	Course Title: Mechanism of Printing Machines II
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1 P:0)	Periods per semester: 60

Course Objectives:

- To provide a clear and sound knowledge in some of the major print production systems
- To make judgment about the aspect of printing, differentiate each process.
- To impart knowledge so that the students can identify the selection of a particular process to choose for a specific print production.
- To impart knowledge so that the students can identify the problems that occurs during the printing process.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Parts of machine		Mechanism of printing machine I	1
Cylinder structure		Mechanism of printing machine I	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the flexographic printing process.	15	Understanding
CO2	Explain comprehend the Gravure printing process.	14	Understanding
CO3	Explain comprehend the Screen-printing process.	15	Understanding
CO4	Assess problems, analyze and give remedies.	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the flexographic printing process.		
M1.01	Introduction to Flexography presses.	3	Understanding
M1.02	Image carrier preparations.	4	Understanding
M1.03	Flexo Printing unit.	4	Understanding
M1.04	Flexo substrates.	4	Understanding
Contents: Principles of Flexography Printing Process: Advantages, Limitations and Characteristics. Main Sections of Flexo Printing Machine: Infeed section, Printing section - Inking Unit, Plate carrier, Impression Cylinder, Drying section and outfeed section. Configurations of Flexo- Inline, stack, central impression. Flexo Image Carrier Preparation: Structure of Flexographic Plate/Image carrier - Metal backed plates, Magnetic plates Types of Plates- rubber and photopolymer. Plate Preparation Methods - Rubber Plates preparation, Sheet Photopolymer Plates preparation and Liquid Photopolymer. Flexo Inking Systems: Ink Metering, Anilox Roller. Types of Flexo Inking systems - Two-roll ink metering systems, Modified Two-roll with doctor blade ink metering system, Reverse angle doctor blade ink metering system and Chambered blade ink metering system. Plates Preparation. Anilox Roll specifications - Cell count, Cell depth, Cell volume. Types of Anilox roll based on cell shapes - Inverted Pyramid shape cells, Quadrangular shapes cell and Tri-helical shape cells. Substrates for Flexographic printing - Paper and Paperboard stocks, Corrugated stocks, Plastic Films, Foils and Laminates. Describe Briefly- Corona Treatment. Wide web presses. Corrugated presses. Narrow web press. Future of flexography.			
CO2	Explain comprehend the Gravure printing process.		
M2.01	Introduction to gravure printing process.	3	Understanding
M2.02	Structure of gravure cylinder.	2	Understanding
M2.03	Image carrier preparations and printing unit	5	Understanding
M2.04	Drying chamber and substrates.	4	Understanding
	Series Test I	1	

Contents:

Principles of Gravure Printing Process: Advantages, Limitations and Characteristics of Gravure Process.

Main Sections of Gravure Printing Machine: infeed section, Printing section, Drying section and outfeed section.

Structure of Gravure Cylinder: Gravure cylinder parts - Axis, Shaft, Diameter, Circumference and Face length. Balancing- Static and Dynamic, distinguish static and dynamic balancing. Gravure cylinder well configuration and its advantages and disadvantages.

Gravure Image Carrier Preparation: Gravure Cylinder manufacture -Thin layer method, Ballard skin method and Heavy Copper plating method. Gravure Cylinder Preparation Methods - Conventional Method / Carbon Tissue Method, Electromechanical Engraving method and Laser Engraving method. Doctor Blade - Structure, Types - Conventional doctor blades, MDC / Ringier doctor blades, Counter face doctor blades, Rounded doctor blades,, Mechanisms of doctor blade. Doctor blade wear - Types. Doctor blade materials. Doctor blade problems. Pre-wipe blade. Impression Roller - Structure, functions and covering. Electrostatic assist. Nib width Drying System - Drying Chamber - Solvent Recovery Systems - Environmental Friendly Solvent Removal Systems. Substrates for printing - Paper and non paper.

CO3	Explain comprehend the Screen-printing process.		
M3.01	Introduction to screen printing	3	Understanding
M3.02	Types of stencils	4	Understanding
M3.03	Stencil preparations and substrates	4	Understanding
M3.04	Screen printing machines and applications	4	Understanding

Contents:

Screen printing-definition. Equipment and materials: frames (wooden frames, metal frames, self stretching frames). Mesh materials (silk, synthetic, stainless steel). Types of stencils - Hand-cut, tusche and glue, photographic. Screen fabrics-multifilament screens, monofilament screens. Fabric stretching techniques-mechanical stretching, hand stretching. Introduction to - Photographic stencil methods-direct image method, indirect/transfer image method, film emulsion or direct/indirect image method. Substrates for printing - (paper, paper board, wood, textiles, plastics, metals, ceramics & glass). Image transfer squeegee (types and selection), on contact and off contact printing. Printing process - Manual, semi automatic, fully automatic. Screen printing ink. Definition of Screen reclamation. Applications of screen printing process. Advanced screen printing machines.

CO4	Assess problems, analyze and give remedies.		
M4.01	Selection of printing process according to the job.	5	Understanding
M4.02	Trouble shooting related to flexo.	4	Understanding
M4.03	Trouble shooting related to gravure and screen printing.	5	Understanding
	Series Test II	1	

Contents:

Selection of printing process according to the job-procedures to be followed, workflow in a Printing press, workflow diagram. Trouble shooting related to flexo - tracking, trapping, mottle, striations, weak color, adhesion, foaming, smearing etc. Trouble shooting related to gravure - abrasion, scumming, drying in, bleeding, haze, pinholes, blocking, static, drying in, brittleness, etc. Troubles related to screen printing.

Text / Reference:

T/R	Book Title/Author
T1	Gravure Process and Technology , GAA.
R1	Flexographic Principles and Practices, 4th edition & 5th edition , Foundation of flexographic Technical Association.
R2	Handbook of Printing and Production , Michael Bernard, John Peacock.
R3	Printing Technology 5th edition , Adams J.M, Delmar Publishers, NewYork.
R4	Hand book of print media , Helmut Kipphan, Springer Science & Business Media.

Online resources:

Sl.No	Website Link
1	www.gravurecylinders.co.za/the-process
2	https://www.thebookdesigner.com
3	https://www.colormatters.com
4	https://www.britannica.com/technology/flexography
5	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner
6	https://www.tech-sleeves.com/the-company/flexography